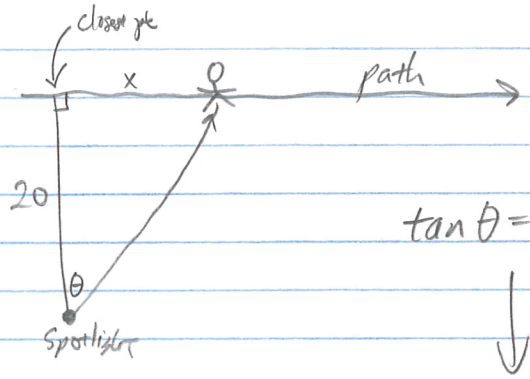
④



$$\tan\theta = \frac{x}{20}$$

$$\downarrow \frac{d}{dt}$$

$$\frac{d}{dt}[\tan\theta] = \frac{d}{dt}\left[\frac{x}{20}\right]$$

$$(\sec^2\theta)\cdot\frac{d\theta}{dt} = \frac{1}{20}\cdot\frac{dx}{dt}$$

$x = 15. \rightsquigarrow \theta = \tan^{-1}\left(\frac{15}{20}\right)$

$$\sec^2\left(\tan^{-1}\left(\frac{15}{20}\right)\right)\cdot\frac{d\theta}{dt} = \frac{1}{20}\cdot 4$$

$$\frac{d\theta}{dt} = \frac{1}{5\sec^2\left(\tan^{-1}\left(\frac{15}{20}\right)\right)} = 0.128$$

Positive ...

The angle θ is increasing at a rate of 0.128 radians/sec.